MANDAR MHAPSEKAR

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Education

Degree	Institute / School	Year	Score
Master of Science in Computer Science	Viterbi School of Engineering (University of Southern California)	2024	3.85/4
Bachelor of Engg. in Information Tech.	D. J. Sanghvi College of Engineering (University of Mumbai)	2019	9.30/10

Work Experience

Software Development Engineer II, JP Morgan Chase & Co **Jan'22 – Oct`22**

- Took complete ownership of the design and development of a payment clearing service project for India's real-time payments and was **smoothly delivered in the span of 6 months**.
- Incorporated hybrid RSA & AES encryption which helped build a secure communication channel with the clearing house.
- Setup the CI/CD pipeline for the project using Jenkins making deployment faster and more efficient **by reducing the turnaround time by 2 hours**.
- Contributed also to the development of a highly configurable, scalable complex payment processing system containing more than ten Microservices that serviced real-time payments in multiple markets worldwide.
- Responsible for **code reviews** and ensuring that **coding design patterns** were up to the team's standard.
- Collaborated with the team on a proof of concept for migrating a payment processing system to the firm's internal Kubernetes platform, which assisted them in understanding the application's performance challenges on the Kubernetes platform.

Software Development Engineer I, JP Morgan Chase & Co **Jul'19 - Dec'21**

- Developed a generic virtual assistant framework leveraging RASA & Django that automated various processes of the production management team, reducing significant manual toil, **thus saving around 2 hours per employee daily**.
- Trained a model using Doc2Vec sentence embedding to suggest similar historical tickets and also implemented a preprocessor using traditional NLP algorithms to extract key phrases and eliminate stop words.

Projects & Publications

Global Clearing Connector (GC2 - IMPS) **Feb`21 – Dec`22**

- GC2 IMPS is a highly scalable Microservice designed to clear millions of payments handled daily by JPMC's India IMPS payments processor by sending/receiving payment information to the clearing house (NPCI).
- The connectivity of this application with the above 3rd party systems is channeled through Apache Kafka, HTTP, and SFTP File.
- Developed a Kafka monitoring service to ensure high resiliency by performing on-the-fly failover to the secondary cluster thus **bringing down the production incidents to 0** caused due to data-center failure on Kafka cluster.
- Implemented the event sourcing design pattern, in which the payment stages were used as events to trigger different Java beans to perform a series of tasks, **resulting in improved project structure and code re-usability**.
- Integrated Token-based authentication to achieve security for end-to-end communication between UI and backend REST APIs.
- Worked on highly complex reconciliation processing that used Apache Camel to process files received from the clearing house and generate a report for the operations team.

Implementation of ResNet for Devanagari Handwritten Character Recognition **Aug'18 - Jul'19**

- Experimented with different deep CNN architectures like ResNet, AlexNet, VGG, Inception for recognizing the main body of a Devanagari handwritten character leveraging transfer learning and got the highest accuracy with the ResNet 50 architecture.
- Accomplished multi-processing and deployed the solution on Google's Compute Engine to **improve the performance by 30%**.
- Published a research paper on this project in the 2nd ICACTA conference – Springer. (Mandar Mhapsekar - Google Scholar)

Honors & Awards

- Awarded the **TCS Best Student Project Award** for the project "Document analysis and handwritten OCR for Devanagari script".
- Won the **Dipex 2019** - state level project competition organized by SRIJAN - DIPEX Center for Entrepreneurship Development.
- Among the top 9 finalists of **AICTE's National Level Chhatra Vishwakarma Awards '17**.
- Secured the runner-up position at **Smart Mobility Hackathon (national level hackathon)** organized by HERE Technologies.
- Won the 2nd Prize at **Smart India Hackathon** for the project "Helping Hands".
- Runner up at VJTI's Technovanza 2018 **m-Indicator hackathon** conducted by m-Indicator.
- Was among the **Top 5% Performing SEPs at JPMorgan** (Entry Level Analyst) from the 2019 batch consisting of 300 analysts.

Skills

Languages: C, Java, JavaScript, Python, SQL,

Technologies: Java Spring Boot, Python Django, React JS., Apache Kafka, Oracle DB, MySQL DB, GemFire Cache, RASA Framework, Android, Kubernetes, Spring Cloud, Eureka, Apache Camel, RASA, Junit, Jest, C24 Transformation, Cloud Foundry, Kerberos Auth.